TECH YUVA × GOOGLE DEVELOPER GROUPS GURUGRAM

BUILD WITH AI

BUILD-A-THON

2026

PROBLEM STATEMENTS

64 challenges • Choose a domain • Build a working prototype

24 OCTOBER Event Date	**9:00 AM – 5:00 PM** 8-hour build-a-thon
THOUGHTWORKS, GURUGRAM Venue	**2–4** Participants per team

INTERNSHIP OPPORTUNITIES • FREE MEALS • PARTICIPANT CREDITS • EXCITING PRIZES

Build a useful software prototype and demonstrate it live.
AI Studio, Google Stitch and Antigravity can support your workflow and interface.

Use sample or authorized data. Include human review for high-impact outputs.

ARTIFICIAL INTELLIGENCE

9 RELATED CHALLENGES

PS-01 | Document-Grounded AI Research Assistant

CONTEXT & PROBLEM

Students and teams often spend too much time searching across long documents and may receive answers without clear evidence.

EXPECTED SOLUTION

Build an assistant that accepts multiple documents, answers questions using their content, cites the relevant passages and indicates when the answer cannot be found in the supplied material.

TECHNICAL FOCUS Document parsing • Retrieval • Source-grounded answers

PS-02 | AI Meeting Summary and Action Tracker

CONTEXT & PROBLEM

Meeting decisions and assigned tasks can be lost when notes are incomplete or scattered across chat messages.

EXPECTED SOLUTION

Convert a supplied transcript into a summary, decisions, task owners and deadlines. Provide an editable action list and a status dashboard.

TECHNICAL FOCUS Text extraction • Structured output • Task workflow

PS-03 | Smart Complaint Classification

CONTEXT & PROBLEM

Organizations receive complaints that may be miscategorized, duplicated or sent to the wrong department.

EXPECTED SOLUTION

Classify reports by topic and urgency, group similar complaints and recommend a department with a short explanation for each recommendation.

TECHNICAL FOCUS Text classification • Similarity matching • Routing

PS-04 | Personalized Learning Recommender

CONTEXT & PROBLEM

Students may keep revising topics they already understand while missing concepts that need more practice.

EXPECTED SOLUTION

Use quiz scores and topic tags to identify learning gaps and recommend targeted exercises. Update recommendations when new results are recorded.

TECHNICAL FOCUS Recommendation logic • Learner profiles • Progress analytics

PS-05 | AI Answer Verification Assistant

CONTEXT & PROBLEM

AI-generated answers may include claims that are not supported by the provided reference material.

EXPECTED SOLUTION

Compare an answer with supplied documents, highlight claims that have supporting evidence and flag claims that need manual verification.

TECHNICAL FOCUS Claim extraction • Evidence matching • Confidence labels

PS-06 | Computer Vision for Safety Alerts and Warehouse Inventory

CONTEXT & PROBLEM

Safety teams may need alerts when a person appears to fall or remain inactive, while warehouses need a faster way to count and locate stored items.

EXPECTED SOLUTION

Build one focused computer-vision demo using sample images or video: either flag selected pose/activity patterns for human review, or detect and count warehouse objects in a defined scene. Show confidence scores and allow manual correction.

TECHNICAL FOCUS OpenCV or pretrained detector • Pose/object detection • Event alerts • Human review

PS-33 | Customer Churn Early-Warning Dashboard

CONTEXT & PROBLEM

Subscription businesses may notice customer disengagement only after a user stops using the product or cancels a subscription.

EXPECTED SOLUTION

Analyze synthetic customer activity, support-ticket and usage data to calculate an explainable churn-risk indicator. Show risk trends and recommend review actions such as a check-in or onboarding guide; do not claim predictions are certain.

TECHNICAL FOCUS Event analytics • Risk scoring • Dashboard • Workflow recommendations

PS-41 | AI-Assisted Sports Movement Feedback

CONTEXT & PROBLEM

Amateur athletes may not have access to specialist motion-analysis facilities to review movement technique.

EXPECTED SOLUTION

Build a web app that accepts a short sample sports-movement video, estimates key body landmarks and displays selected joint angles or left-right symmetry indicators. Present feedback as educational guidance, not a medical diagnosis or guaranteed injury prediction.

TECHNICAL FOCUS Pose estimation • MediaPipe concepts • Angle calculation • Video visualization

PS-64 | Adaptive Skills and Career Competency Platform

CONTEXT & PROBLEM

Students can find it difficult to connect their current skills with the competencies required for a chosen career or internship.

EXPECTED SOLUTION

Build a skill assessment and roadmap tool that maps quiz results to a simple competency graph, identifies prerequisite gaps and recommends project-based learning modules. Update recommendations as assessment results change.

TECHNICAL FOCUS Skill graph • Adaptive assessment • Recommendation logic • Progress tracking CYBERSECURITY 6 related challenges

CYBERSECURITY

6 RELATED CHALLENGES

PS-07 | Phishing Link Risk Analyzer

CONTEXT & PROBLEM

Users may struggle to recognize suspicious links before opening them.

EXPECTED SOLUTION

Analyze URL features and a supplied list of known indicators to produce a risk score, highlight warning signs and provide safe next-step guidance without visiting suspicious sites.

TECHNICAL FOCUS URL analysis • Risk scoring • Explainable results

PS-08 | Login Log Anomaly Detector

CONTEXT & PROBLEM

Repeated failed logins and unusual access patterns can be difficult to spot in large log files.

EXPECTED SOLUTION

Process sample authentication logs, identify abnormal login times and repeated failures, and display alerts with the relevant log evidence.

TECHNICAL FOCUS Log parsing • Baseline rules • Anomaly detection

PS-09 | Software Dependency Vulnerability Checker

CONTEXT & PROBLEM

Applications can depend on packages with known security issues, but developers may not know which versions need attention.

EXPECTED SOLUTION

Read a sample dependency file, match package versions against a vulnerability dataset and generate a prioritized list of findings with upgrade guidance.

TECHNICAL FOCUS Dependency parsing • Vulnerability matching • Prioritization

PS-10 | Secret Detection in Code

CONTEXT & PROBLEM

Developers can accidentally include API keys or other secrets in source code and configuration files.

EXPECTED SOLUTION

Scan a provided sample project for likely exposed secrets, mask the values in the results and produce a report explaining safe remediation steps.

TECHNICAL FOCUS Pattern matching • False-positive handling • Secure reporting

PS-11 | Digital Evidence Integrity Tracker

CONTEXT & PROBLEM

Digital evidence needs consistent records so that investigators can check whether files have changed after collection.

EXPECTED SOLUTION

Create a case workspace that records file hashes, timestamps, evidence notes and chain-of-custody events, with a verification feature for sample files.

TECHNICAL FOCUS Hashing • Audit logs • Evidence tracking

PS-12 | Web Security Auditor for AI-Generated Applications

CONTEXT & PROBLEM

Web applications created quickly with AI-assisted coding may contain exposed endpoints, weak input validation or missing access checks.

EXPECTED SOLUTION

Build an authorized, non-destructive audit tool for a local demo application or supplied sample code. Check selected security misconfigurations, show evidence and severity, and provide practical remediation guidance.

TECHNICAL FOCUS OWASP concepts • Static checks • Endpoint inventory • Remediation reporting BLOCKCHAIN & WEB3 5 related challenges

BLOCKCHAIN & WEB3

5 RELATED CHALLENGES

PS-13 | Crypto Transaction Relationship Visualizer

CONTEXT & PROBLEM

Understanding the movement of cryptocurrency between many wallet addresses can be difficult from raw transaction lists.

EXPECTED SOLUTION

Visualize a supplied transaction dataset as a graph, highlight repeated transfers and unusual flow patterns, and let users inspect the transactions behind each connection.

TECHNICAL FOCUS Graph visualization • Transaction analysis • Risk indicators

PS-14 | Digital Certificate Verification

CONTEXT & PROBLEM

Institutions need a reliable way to check whether a digital certificate is genuine and has not been altered.

EXPECTED SOLUTION

Build a prototype that records certificate fingerprints, verifies submitted certificates and supports a revoked-certificate status using a test network or tamper-evident database.

TECHNICAL FOCUS Hashing or signatures • Verification • Status checks

PS-15 | Transparent Donation Tracker

CONTEXT & PROBLEM

Donors may not have a clear view of how contributions are assigned to projects and recorded expenses.

EXPECTED SOLUTION

Create a contribution and expense ledger with project-wise summaries, transaction history and alerts when recorded totals do not reconcile.

TECHNICAL FOCUS Ledger design • Reconciliation • Dashboard

PS-16 | QR-Based Medicine Traceability

CONTEXT & PROBLEM

Customers and pharmacies need a way to check the recorded journey of a medicine through the supply chain.

EXPECTED SOLUTION

Generate QR codes for sample product batches and display handoff records from manufacturer to distributor and retailer. Flag missing or inconsistent steps.

TECHNICAL FOCUS QR codes • Traceability • Data validation

PS-17 | Consent-Based Digital Credential Wallet

CONTEXT & PROBLEM

Users may need to share proof of a qualification without sharing an entire identity document.

EXPECTED SOLUTION

Build a prototype that stores signed sample credentials and lets users share selected information with a verifier after reviewing a consent screen.

TECHNICAL FOCUS Digital signatures • Consent workflow • Data minimization DATA SCIENCE & ANALYTICS 7 related challenges

DATA SCIENCE & ANALYTICS

7 RELATED CHALLENGES

PS-18 | Student Engagement Insights

CONTEXT & PROBLEM

Educators may notice student disengagement only after assignments and attendance have declined.

EXPECTED SOLUTION

Analyze anonymized sample data to find engagement patterns, visualize contributing factors and provide cautious, explainable risk indicators for human review.

TECHNICAL FOCUS Data analysis • Classification basics • Explainability

PS-19 | Traffic Demand Forecast Dashboard

CONTEXT & PROBLEM

Traffic levels vary by time, weekday and weather, making travel planning difficult.

EXPECTED SOLUTION

Use historical or sample traffic data to forecast demand for selected time periods and compare the results with a simple baseline model.

TECHNICAL FOCUS Time-series forecasting • Feature engineering • Model comparison

PS-20 | Unusual Transaction Detector

CONTEXT & PROBLEM

Large datasets may contain rare transaction patterns that deserve closer review.

EXPECTED SOLUTION

Analyze a synthetic transaction dataset, flag unusual records and show the features that contributed to each alert. Present alerts as indicators, not proof of fraud.

TECHNICAL FOCUS Anomaly detection • Data preparation • Explainable alerts

PS-21 | Electricity Usage Forecast

CONTEXT & PROBLEM

Campuses and small buildings may not know when electricity demand is likely to peak.

EXPECTED SOLUTION

Forecast short-term consumption from sample historical data, show expected peak periods and compare predicted values with actual readings.

TECHNICAL FOCUS Forecasting • Data visualization • Error measurement

PS-22 | Public Transport Demand Analyzer

CONTEXT & PROBLEM

Transport planners need to know which routes and time periods experience the highest passenger demand.

EXPECTED SOLUTION

Analyze route and passenger-count data, visualize demand patterns and recommend where service frequency could be reviewed.

TECHNICAL FOCUS Data cleaning • Trend analysis • Decision support

PS-51 | Air Quality Trend Explorer

CONTEXT & PROBLEM

Air pollution data is harder to interpret without comparing time, weather and location.

EXPECTED SOLUTION

Combine public or sample air-quality and weather datasets to visualize trends and compare patterns across locations.

TECHNICAL FOCUS Data integration • Visualization • Trend analysis

PS-57 | Appointment No-Show Insights

CONTEXT & PROBLEM

Missed appointments reduce service capacity and can delay access to care.

EXPECTED SOLUTION

Use synthetic appointment records to identify patterns associated with missed visits and suggest reminder strategies for review.

TECHNICAL FOCUS Data analysis • Classification basics • Responsible use PRODUCT DEVELOPMENT 8 related challenges

PRODUCT DEVELOPMENT

8 RELATED CHALLENGES

PS-28 | Customer Feedback-to-Feature Planner

CONTEXT & PROBLEM

Product teams may receive lots of feedback but struggle to decide which improvements matter most.

EXPECTED SOLUTION

Group sample reviews by topic, identify recurring problems and produce a prioritized feature list that teams can edit and track.

TECHNICAL FOCUS Text analytics • Topic grouping • Prioritization

PS-29 | Smart Inventory and Reorder Assistant

CONTEXT & PROBLEM

Small businesses can lose sales through stockouts or waste money by ordering excess inventory.

EXPECTED SOLUTION

Use sample sales and stock data to estimate demand, flag low-stock items and recommend reorder quantities with visible assumptions.

TECHNICAL FOCUS Forecasting basics • Inventory rules • Dashboard

PS-30 | Product Impact Comparison Tool

CONTEXT & PROBLEM

Customers and product teams need to compare the environmental impact of alternative product choices.

EXPECTED SOLUTION

Build a calculator that compares material, transport and energy assumptions across products and displays which inputs most affect the estimate.

TECHNICAL FOCUS Scenario analysis • Calculation engine • Visualization

PS-31 | Customer Retention Insights Dashboard

CONTEXT & PROBLEM

Businesses may lose customers without noticing changes in activity or satisfaction.

EXPECTED SOLUTION

Analyze sample customer activity, identify users showing reduced engagement and recommend follow-up actions with clear explanations.

TECHNICAL FOCUS Segmentation • Trend analysis • Rule-based scoring

PS-32 | Website Accessibility Audit Assistant

CONTEXT & PROBLEM

Web pages can exclude users when keyboard navigation, labels or color contrast are not handled properly.

EXPECTED SOLUTION

Check a sample web page against selected accessibility rules, list issues with evidence and provide practical suggestions for fixes.

TECHNICAL FOCUS Accessibility checks • Issue reporting • Remediation

PS-34 | Smart POS and Billing for Cafes

CONTEXT & PROBLEM

Small cafes may struggle with table tracking, split bills, kitchen order communication and inventory updates during busy periods.

EXPECTED SOLUTION

Build a POS prototype with a visual table layout, menu and order management, split-billing, kitchen order tickets and stock deduction. Simulate QR payment confirmation and demonstrate how orders can be queued during a temporary offline state.

TECHNICAL FOCUS Web app • Table and order workflow • KOT • Inventory logic • Offline simulation

PS-42 | Personal Budget Scenario Planner

CONTEXT & PROBLEM

People with changing income may struggle to understand how spending choices affect their monthly budget.

EXPECTED SOLUTION

Categorize sample income and expenses, compare savings scenarios and show how changes in spending affect a planned budget without connecting to real financial accounts.

TECHNICAL FOCUS Categorization • Scenario modeling • Visualization

PS-55 | Emergency Queue Flow Simulator

CONTEXT & PROBLEM

Crowded emergency departments need visibility into queue status and possible bottlenecks while clinical staff retain triage authority.

EXPECTED SOLUTION

Simulate patient flow through departments, estimate waiting times and highlight bottlenecks using staff-defined priority levels.

TECHNICAL FOCUS Queue simulation • Workflow visualization • Bottleneck analysis REAL-WORLD & CIVIC TECH 5 related challenges

REAL-WORLD & CIVIC TECH

5 RELATED CHALLENGES

PS-23 | Civic Complaint Tracking Platform

CONTEXT & PROBLEM

Reports about waste, damaged roads and streetlights can be duplicated or remain unresolved without clear status updates.

EXPECTED SOLUTION

Build a portal that categorizes reports, groups likely duplicates, assigns them to a simulated department and tracks resolution status with evidence.

TECHNICAL FOCUS Workflow design • Similarity matching • Status tracking

PS-24 | Smart Queue and Appointment Manager

CONTEXT & PROBLEM

Service counters may experience long waits when demand changes and appointments are cancelled.

EXPECTED SOLUTION

Create a token and appointment system that estimates waiting times, handles cancellations and offers newly available slots to users.

TECHNICAL FOCUS Scheduling • Queue simulation • Notifications

PS-25 | Emergency Resource Coordination Board

CONTEXT & PROBLEM

During a local emergency, requests and available supplies may be spread across disconnected lists.

EXPECTED SOLUTION

Use simulated incident data to match requests with available resources, prioritize urgent needs and track dispatch status on a shared dashboard.

TECHNICAL FOCUS Resource matching • Priority rules • Dashboard

PS-26 | Accessible Route Comparison Tool

CONTEXT & PROBLEM

The shortest route may not be practical for people who need step-free access or accessible facilities.

EXPECTED SOLUTION

Build a route comparison prototype that uses available accessibility details, walking distance and transport options to explain the trade-offs between routes.

TECHNICAL FOCUS Route data • Multi-criteria scoring • Accessibility

PS-27 | Local Service Matching Platform

CONTEXT & PROBLEM

People may find it difficult to identify local providers who match their needs, skills and availability.

EXPECTED SOLUTION

Match service requests with provider profiles using skills, availability, location and feedback, and explain why each result was recommended.

TECHNICAL FOCUS Ranking • Matching logic • Feedback HEALTHCARE & HOSPITALS 9 related challenges

HEALTHCARE & HOSPITALS

9 RELATED CHALLENGES

PS-35 | Patient Follow-up Reminder System

CONTEXT & PROBLEM

Patients may miss follow-up appointments when schedules and reminders are not managed in one place.

EXPECTED SOLUTION

Create a dashboard for follow-up dates, reminder status and missed appointments using synthetic patient records and role-based access.

TECHNICAL FOCUS Scheduling • Access control • Status tracking

PS-36 | Medical Report Trend Viewer

CONTEXT & PROBLEM

People may find it difficult to understand how their laboratory results change over time.

EXPECTED SOLUTION

Display sample lab values over time and compare them with reference ranges supplied with the report, while making clear that clinical interpretation belongs to a healthcare professional.

TECHNICAL FOCUS Data visualization • Trend analysis • Safe communication

PS-37 | Hospital Readmission Pattern Analysis

CONTEXT & PROBLEM

Healthcare planners need to understand which operational or demographic patterns are associated with repeat hospital visits.

EXPECTED SOLUTION

Analyze synthetic records, visualize associated factors and explain limitations and uncertainty rather than making clinical decisions.

TECHNICAL FOCUS Data analysis • Classification • Responsible interpretation

PS-38 | Medication Interaction Reference Checker

CONTEXT & PROBLEM

Patients and caregivers need accessible ways to check medicines against reliable reference information.

EXPECTED SOLUTION

Compare entered medicines against a trusted supplied interaction dataset, flag possible matches and direct users to a pharmacist or clinician for review.

TECHNICAL FOCUS Reference lookup • Rule matching • Safety escalation

PS-39 | Accessible Care Instructions Assistant

CONTEXT & PROBLEM

Complex care instructions may be difficult to follow for people who prefer simpler language or different accessibility options.

EXPECTED SOLUTION

Convert clinician-approved sample instructions into plain-language checklists with multilingual display and text-to-speech support.

TECHNICAL FOCUS Text simplification • Translation workflow • Accessibility

PS-40 | Consent-Based Emergency Patient Record Bridge

CONTEXT & PROBLEM

In an emergency, authorized responders may not immediately have access to relevant patient information, especially across separate healthcare systems.

EXPECTED SOLUTION

Create a prototype using synthetic patient records that exposes selected emergency details through a time-limited QR token or simulated access code. Include consent and access logging, expiration and revocation controls.

TECHNICAL FOCUS Structured health data • QR/token workflow • Access control • Audit logs

PS-54 | Hospital Bed Availability Dashboard

CONTEXT & PROBLEM

Hospital teams need a shared view of bed availability and resource status to coordinate operational decisions.

EXPECTED SOLUTION

Build a dashboard with simulated beds and resources, staff-managed allocation, status changes and a basic audit history.

TECHNICAL FOCUS Resource tracking • Role-based workflow • Audit logs

PS-56 | Medical Inventory Expiry Monitor

CONTEXT & PROBLEM

Hospitals must monitor stock quantities and expiry dates to reduce waste and prevent shortages.

EXPECTED SOLUTION

Track sample inventory batches, flag near-expiry items, estimate reorder needs and generate a prioritized procurement list.

TECHNICAL FOCUS Inventory management • Expiry tracking • Reorder logic

PS-58 | Hospital Operations Dashboard

CONTEXT & PROBLEM

Managers often need to combine waiting times, appointment use, inventory and feedback to identify operational issues.

EXPECTED SOLUTION

Build a unified dashboard with department-level indicators, date filters and alerts for areas that need human review.

TECHNICAL FOCUS KPI design • Dashboard integration • Trend analysis EDUCATION & EDTECH 5 related challenges

EDUCATION & EDTECH

5 RELATED CHALLENGES

PS-59 | Syllabus-Based Question Paper Generator

CONTEXT & PROBLEM

Teachers spend time preparing balanced assessments across topics, difficulty levels and marks.

EXPECTED SOLUTION

Generate editable questions from supplied study material, group them by topic and difficulty, and apply a configurable marks blueprint with teacher review.

TECHNICAL FOCUS Document processing • Question generation • Assessment rules

PS-60 | Adaptive Learning Recommendation Engine

CONTEXT & PROBLEM

Students with different learning needs often receive the same revision material.

EXPECTED SOLUTION

Use quiz performance and topic tags to recommend targeted practice and update each student's revision plan as new results arrive.

TECHNICAL FOCUS Recommendation logic • Topic mapping • Progress analytics

PS-61 | Internship and Hackathon Matcher

CONTEXT & PROBLEM

Students may miss opportunities because requirements, skills and deadlines are listed across different sources.

EXPECTED SOLUTION

Match student-selected skills and eligibility with a curated list of opportunities, explain skill gaps and display deadlines and source links.

TECHNICAL FOCUS Ranking • Eligibility rules • Data normalization

PS-62 | Assignment Similarity Review Tool

CONTEXT & PROBLEM

Educators need to review similarities between submissions without relying on unsupported automatic accusations.

EXPECTED SOLUTION

Compare sample assignments, highlight similar passages and provide a report for human review rather than declaring plagiarism automatically.

TECHNICAL FOCUS Text similarity • Passage highlighting • Review workflow

PS-63 | Classroom and Lab Timetable Optimizer

CONTEXT & PROBLEM

Institutions need to schedule classes, faculty and specialized labs while avoiding room and time conflicts.

EXPECTED SOLUTION

Build a timetable tool that allocates sample classes based on capacity and equipment requirements, detects conflicts and suggests alternative slots.

TECHNICAL FOCUS Scheduling • Constraint rules • Conflict detection LIFE & SOCIAL IMPACT 5 related challenges

LIFE & SOCIAL IMPACT

5 RELATED CHALLENGES

PS-43 | Community Food Donation Matcher

CONTEXT & PROBLEM

Surplus food and community needs may coexist because donors and recipient organizations cannot coordinate quickly.

EXPECTED SOLUTION

Match sample food donations with registered recipient organizations by quantity, location and collection window, with expiry and collection-status tracking.

TECHNICAL FOCUS Matching • Scheduling • Inventory status

PS-44 | Accessibility Barrier Reporting Map

CONTEXT & PROBLEM

Reports about inaccessible public facilities are often scattered across different channels.

EXPECTED SOLUTION

Create a map-based portal to submit barrier reports, group duplicates, let moderators verify entries and track whether issues are resolved.

TECHNICAL FOCUS Geospatial interface • Moderation • Duplicate detection

PS-45 | Public Scheme Eligibility Guide

CONTEXT & PROBLEM

People may struggle to interpret eligibility criteria and find the correct official application process for public support schemes.

EXPECTED SOLUTION

Create a rules-based guide that asks relevant questions, explains potential matches and links to official scheme information for confirmation.

TECHNICAL FOCUS Rules engine • Form workflow • Source linking

PS-46 | Volunteer and Community Task Matcher

CONTEXT & PROBLEM

Community organizations need to match volunteer skills and availability with tasks that require support.

EXPECTED SOLUTION

Recommend volunteers for tasks based on skills, schedule and location, then track assignment, confirmation and completion.

TECHNICAL FOCUS Matching logic • Scheduling • Workflow

PS-47 | Micro-Savings Planner for Gig Workers

CONTEXT & PROBLEM

Freelancers and gig workers often receive irregular income, making fixed monthly savings plans difficult to follow.

EXPECTED SOLUTION

Create a budgeting prototype that accepts simulated earnings and expenses, forecasts short-term cash flow and suggests adjustable small savings amounts while preserving a user-defined minimum balance. Use mock data; do not move real money.

TECHNICAL FOCUS Cash-flow forecasting • Budget rules • Scenario simulation • Financial dashboard CLIMATE & SUSTAINABILITY 5 related challenges

CLIMATE & SUSTAINABILITY

5 RELATED CHALLENGES

PS-48 | Local Flood Awareness Dashboard

CONTEXT & PROBLEM

Residents need understandable views of rainfall and historical flood information while recognizing that indicators do not guarantee a flood forecast.

EXPECTED SOLUTION

Visualize public or sample rainfall and flood-history data, compare locations and show data dates, sources and limitations.

TECHNICAL FOCUS Data integration • Geospatial visualization • Uncertainty

PS-49 | Waste Collection Route Planner

CONTEXT & PROBLEM

Collection teams may waste time when pickup routes do not reflect current bin status or collection demand.

EXPECTED SOLUTION

Use simulated bin-fill levels and location data to prioritize pickups and compare an optimized route with a simple baseline.

TECHNICAL FOCUS Route planning • Priority scoring • Map dashboard

PS-50 | Campus Energy Anomaly Monitor

CONTEXT & PROBLEM

Unusual electricity use may go unnoticed when buildings only review monthly totals.

EXPECTED SOLUTION

Analyze sample meter readings, flag unusual usage periods and estimate potential savings using clearly stated assumptions.

TECHNICAL FOCUS Time-series analysis • Anomaly detection • Scenario estimates

PS-52 | Water Leak Prioritization Tool

CONTEXT & PROBLEM

Maintenance teams need to prioritize potential leaks when sensor readings and resident reports arrive together.

EXPECTED SOLUTION

Analyze simulated flow readings and issue reports, rank inspection tasks and display the evidence behind each priority.

TECHNICAL FOCUS Anomaly detection • Rule-based scoring • Task tracking

PS-53 | EV Intelligence and Battery Optimization Simulator

CONTEXT & PROBLEM

Electric vehicles need software that balances battery usage, changing road conditions and safety-related events. Testing every driving scenario on a physical vehicle is not practical for a student build-a-thon.

EXPECTED SOLUTION

Build a software simulator that consumes sample speed, battery temperature, obstacle distance and road-condition inputs. Display telemetry, estimate energy usage, recommend safe driving parameters and simulate fault alerts or recovery states.

TECHNICAL FOCUS Python or C++ simulation • Telemetry dashboard • Battery logic • Path-planning concepts